

## EXECUTIVE SUMMARY

Technical expert in Python having more than 2 years of industry experience. Working in the LifeScience domain. Skilled in AI/ML Development, Power BI, Python libraries and AI/ML Algorithms. Effective collaborator and problem-solver.

## FUNCTIONAL EXPERTISE

- Python Development
- AI Development
- Machine Learning
- Data Analysis
- Data Visualization
- Cloud Solutions

## EXPERIENCE

[TATA Consultancy Services](#) (August 2022 - Present)

Working for Client [GlaxoSmithKline Pharmaceuticals \(GSK\)](#) on the below projects:

### Python Automation Engineer

Working as Python Automation Engineer to develop a tool for migrating 200+ of pharmaceutical studies from legacy platforms to a new advanced system.

- Designing and developing a robust framework using Python capable of handling large amounts of data.
- Developing Python scripts to automatically generate log reports and identify any errors during the migration process.
- Inventory and Globally Unique Identifier creation in Microsoft SQL Server using SQL for each study to track the migration process.
- Collaborating with stakeholders to understand the requirements and objectives of the migration project.
- Designing and developing automated scripts using Python to extract, transform, and load data from legacy platforms to the new system.
- Working on tools such as [Domino Data Labs](#), [VS Code](#), [PuTTY](#), [SAS Studio](#), [Azure Data Lake Storage](#).
- Optimizing scripts for performance and scalability to handle large volumes of pharmaceutical study data.
- Conducting testing and debugging of scripts to identify and resolve issues throughout the migration process.

## **PowerBi Developer**

- I create dynamic and interactive dashboards that allow stakeholders to intuitively explore and understand complex datasets.
- By connecting Power BI with Azure DevOps, I seamlessly integrate project management data with performance metrics, enabling teams to gain holistic insights into project progress, resource allocation, and overall efficiency.
- I implement automated data refresh schedules and leverage Azure DevOps APIs to ensure real-time updates, providing teams with accurate and up-to-date information for timely decision-making.
- My technical expertise extends to configuring Power BI gateways and managing data access permissions, ensuring secure and reliable data delivery to stakeholders.

## **PROOF OF CONCEPTS**

(Link: <https://github.com/amoljain2k>)

### **NLP Chatbot using GCP Dialogflow, FastAPI and MySQL (FullStack)**

- The Food Ordering Chatbot is an intelligent conversational agent designed to streamline the food ordering process for a food delivery website built using Python, Dialogflow, FastAPI, MySQL, Uvicorn, Ngrok, HTML and CSS.
- Used GCP Dialogflow for natural language understanding and FastAPI for backend integration.
- The chatbot interacts with users, takes their orders, and stores order details in a MySQL database
- Defined Intents and Entities in Dialogflow, intents represent the actions that the chatbot can perform and Entities are data points extracted from user inputs.
- Using Prompt Engineering Intents are trained with various user expressions to ensure accurate understanding and response to diverse queries.

### **AI Content Generator using LLM Model Llama2**

- Developed a web application using Streamlit for generating blog posts.
- This application leverages the Llama 2 large language model (LLM) to produce human-like text based on user inputs.
- The core functionality includes taking user inputs for the blog topic, desired word count, and target audience, and generating a blog post accordingly.
- Utilized Streamlit for the web interface and Langchain libraries for LLM integration.
- Configured the Llama 2 model with specific settings (token limit, temperature).

### **Product Sales and Financial Performance Analysis using Power Bi**

- Aggregated sales data from multiple sources, clean and preprocess the data to ensure accuracy and consistency.
- Create interactive visualizations to allow users to explore data and drill down into specific details.

- Develop multiple pages within the dashboard to cater to different user personas, such as executives, sales managers, and sales representatives.
- Compare regional performance, analyze product-level trends, and identify high-value customers using DAX and M language.

### Password Strength Prediction using Natural Language Processing(NLP)

- Developed a password strength prediction system using NLP techniques, achieving an accuracy rate of 80% in categorizing passwords
- Cleaning the dataset by removing any irrelevant or duplicate entries and extracting relevant features from the passwords that can aid in predicting their strength.
- Utilize NLP techniques to convert string values ML algorithm understandable format vectors.
- Employ machine learning algorithm Logistic Regression to train the predictive model on the preprocessed dataset

## EDUCATION

| Year | Degree       | Institute                                    | CGPA/Percentage |
|------|--------------|----------------------------------------------|-----------------|
| 2022 | B.TECH (CSE) | Moradabad Institute Of Technology, Moradabad | 72.6%           |
| 2018 | CBSE         | Bal Vidya Mandir Sr.Sec.School, Sambhal      | 69.0%           |
| 2016 | CBSE         | St.Mary's School, Sambhal                    | 8.8 CGPA        |

### *Additional Professional Development:*

- Professional Machine Learning Engineer - Google Cloud Platform (2024)
- Power BI Data Analyst Associate(PL-300) - Microsoft (2024)
- Associate Cloud Engineer - Google Cloud Platform (2023)
- Azure Data Engineer Associate(DP-203) - Microsoft (2023)
- Azure Fundamentals(AZ-900) - Microsoft (2022)

## SKILLS

**Languages:** Python, SQL, PL/SQL, DAX, M, Bash Scripts.

**Tools:** Power BI, Domino Data Labs, Jupyter Notebook, SAS Studio, PuTTY, VS Code, Informatica.

**Technical Knowledge:** Machine Learning, Artificial Intelligence, Natural Language

Processing(NLP), Neural Networks, Image Processing, Deep Learning, LLM, Llama 2, OpenAI, FastAPI, GitHub, Git, SQL, Cloud Development, AI Algorithms, Data Analysis, Data Visualization, Database integration, Python Libraries such as Numpy, Pandas, Matplotlib, scikit-learn, Keras, Requests, Beautiful Soup, OpenCV, TensorFlow, Streamlit, Langchain, GCP, AWS, Dialog Flow.